FOOT & ANKLE · NEET-SS / FET / INI-SS

Ankle arthritis and instability

Part A: Ankle instability

1. Lateral ligament complex

- **Anterior talofibular ligament (ATFL): weakest and most commonly injured**; **intracapsular**; **taut in plantarflexion** (resists **anterior translation and inversion in plantarflexion**); from the anterior fibula about **1 cm proximal to the tip** to the talar neck/body.
- **Calcaneofibular ligament (CFL): extracapsular**, **deep to the peroneal tendons (floor of the peroneal sheath)**; **taut in dorsiflexion**; resists **inversion in neutral/dorsiflexion**; **crosses both the ankle and subtalar joints**.
- **Posterior talofibular ligament (PTFL):** strongest; **rarely injured** (only in dislocations).
- **Injury sequence (inversion in plantarflexion): ATFL first, then CFL, then PTFL.**

2. Acute lateral ankle sprain

- **Commonest sports injury**; **inversion + plantarflexion**.
- **Grading: I** stretch, **II** partial tear (ATFL torn), **III** complete tear (ATFL + CFL).
- **Ottawa ankle rules (radiograph if): pain in the malleolar zone AND** (**bone tenderness at the posterior edge or tip of either malleolus (distal 6 cm)**, or **inability to bear weight (4 steps) immediately and in the emergency department**); **foot: navicular or base of 5th metatarsal tenderness** (**sensitivity close to 100%**; reduce radiographs by about 30-40%).

- **Examination (delayed 4-5 days improves accuracy): anterior drawer test** (**ATFL**; **over 10 mm absolute or over 3-5 mm side-to-side difference = abnormal**), **talar tilt test** (**CFL**; **over 10-15 degrees or over 5 degrees side-to-side**).
- **Differential diagnosis/associated injuries: high ankle (syndesmotic) sprain**, **osteochondral lesion of the talus (OLT)**, **peroneal tendon injury**, **fractures: lateral process of the talus ("snowboarder's fracture"), anterior process of the calcaneus, base of the 5th metatarsal, os trigonum**, **Maisonneuve**, **deltoid injury**, **subtalar instability**.
- **Treatment: functional treatment** (**brief protection (brace/boot for severe), early mobilisation, proprioceptive and peroneal strengthening**) **is superior to immobilisation** for most; **grade III: short below-knee cast/boot (about 10 days) improves early outcomes (Lamb, CAST trial)**; **acute surgical repair rarely indicated** (elite athletes debatable).
- **Chronic symptoms after sprains in about 20-40%** (pain, instability).

3. Chronic lateral ankle instability

- **Mechanical instability** (**pathological laxity: positive drawer/tilt**) vs **functional instability** (**subjective giving way: proprioceptive deficit, peroneal weakness**).
- **Evaluate: hindfoot alignment (cavovarus: Coleman block test)**, **generalised laxity (Beighton)**, **peroneal tendons (split tears)**, **OLT**, **anterior impingement**, **subtalar instability**, **syndesmosis**.
- **Imaging: stress radiographs** (**anterior drawer: over 10 mm absolute or 3 mm side-to-side; talar tilt over 10 degrees or 5 degrees side-to-side**), **MRI** (associated lesions), **ultrasound**.
- **Non-operative first: peroneal strengthening and proprioceptive (balance board) rehabilitation for at least 3-6 months**, braces.
- **Surgery:**

  - **Anatomic repair: modified Broström-Gould procedure (gold standard) (imbrication/shortening of the ATFL and CFL + reinforcement with the inferior extensor retinaculum (Gould modification))**: **about 85-95% good results**; **arthroscopic Broström** increasingly used.
  - **Augmentation (suture tape/internal brace)** for poor tissue, hyperlaxity, high-demand.
  - **Anatomic reconstruction with free tendon graft (gracilis/semitendinosus/allograft)**: **failed Broström, poor-quality tissue, generalised laxity (Ehlers-Danlos), heavy patients, long-standing instability**.
  - **Non-anatomic tenodeses (peroneus brevis): Watson-Jones, Evans, Chrisman-Snook**: **historical**: **restrict subtalar motion, stiffness, sacrifice of PB**.
  - **Correct cavovarus (lateralising calcaneal osteotomy, first metatarsal dorsiflexion osteotomy)** if present: **otherwise failure**.
  - **Concomitant ankle arthroscopy** for intra-articular lesions (OLT, impingement).
- **Complications: superficial peroneal nerve injury (commonest)**, **sural nerve injury**, **stiffness (loss of inversion)**, recurrence.

4. Syndesmotic (high ankle) sprains

- **External rotation + dorsiflexion**: **AITFL (first), interosseous ligament, PITFL, transverse ligament**; **collision sports**; **longer recovery (about 2x lateral sprains)**.
- **Tests: squeeze test (Hopkin)**, **external rotation stress test (Kleiger: most sensitive)**, **crossed-leg test**, **Cotton test (lateral translation of the talus)**, **fibular translation test**.
- **Radiographs (weight-bearing/stress): tibiofibular clear space over 6 mm (1 cm above the plafond on AP and mortise: most reliable)**, **tibiofibular overlap under 6 mm on AP (or under 1 mm**

on mortise), **medial clear space over 4-5 mm (deltoid injury)**; **bilateral weight-bearing CT** (subtle diastasis), **MRI**; **arthroscopy (gold standard: over 2 mm diastasis on probing)**.

- **Treatment: stable (no diastasis): non-operative (boot, protected weight bearing)**; **unstable/diastasis: fixation (syndesmotic screws (3.5-4.5 mm, 1-2 screws, 3-4 cortices, about 2-5 cm above the plafond, angled 20-30 degrees anteriorly from posterior) or suture-button devices (dynamic; fewer removals, better reduction))**; **anatomic reduction of the fibula in the incisura is the key (malreduction = poor outcome)**; **direct visualisation/CT** recommended.
- **Chronic syndesmotic instability:** debridement + fixation/reconstruction; **arthrodesis** of the syndesmosis (salvage).

5. Medial (deltoid) ligament injury and instability

- **Superficial deltoid** (tibiospring, tibionavicular, tibiocalcaneal: **resists hindfoot eversion**) and **deep deltoid (anterior and posterior tibiotalar: primary restraint to lateral talar shift and external rotation)**.
- **Isolated deltoid injuries are rare** (usually with **fractures (bimalleolar equivalent)** or syndesmotic injury); **chronic medial instability** associated with **PCFD (stage IV)** and valgus ankles.
- **Gravity stress test / external rotation stress test**: **medial clear space over 4-5 mm** (or **over 1 mm greater than the superior clear space**) = deep deltoid insufficiency.
- **Repair/reconstruction (anatomic, anchors, graft)**.

6. Subtalar instability

- Often **coexists with lateral ankle instability (CFL, cervical and interosseous talocalcaneal ligaments)**; difficult to diagnose (**Brodén stress views**); **reconstruct the CFL** (anatomic reconstruction addressing both joints).

Part B: Ankle arthritis

7. Epidemiology and aetiology

- **Post-traumatic (about 70-80%: rotational ankle fractures, pilon fractures, recurrent instability)**, **primary OA (uncommon: about 10%)**, **inflammatory (RA)**, **haemophilia**, **neuropathic (Charcot)**, **AVN of the talus**, **septic sequelae**, **varus/valgus malalignment (cavovarus, PCFD)**.
- **Ankle cartilage is thinner (about 1-1.7 mm) but more resistant to OA** than the knee (**higher proteoglycan content, stiffer, less response to catabolic cytokines (IL-1), smaller contact area but high congruency**): **younger patients (post-traumatic) than hip/knee OA**; **disability comparable to end-stage hip OA or heart failure (SF-36)**.

8. Evaluation

- **Weight-bearing AP, mortise and lateral ankle + foot radiographs**, **hindfoot alignment view (Saltzman)**, **long-leg alignment**; **weight-bearing CT**.
- **Classifications: Takakura (varus OA: stage 1 sclerosis/osteophytes, 2 medial narrowing, 3A medial gutter obliteration, 3B narrowing to the talar dome, 4 whole joint)**; **Kellgren-Lawrence**; **COFAS** (for TAR planning).
- **Assess: coronal deformity (varus/valgus: congruent vs incongruent)**, **ligament stability**, **hindfoot/subtalar arthritis**, **bone stock**, **soft tissues**, **vascular status**, **neuropathy**, **adjacent joint arthritis**, **smoking**, **infection history**.

9. Non-operative treatment

- **Activity modification, weight loss, NSAIDs, bracing (lace-up brace, AFO (Arizona)), rocker-bottom sole with SACH heel (solid**

ankle cushioned heel), **intra-articular steroid/hyaluronic acid (limited evidence)**, PRP (experimental).

10. Joint-preserving surgery

- **Arthroscopic debridement (anterior impingement osteophytes)**: **for impingement symptoms with preserved joint space**; poor for diffuse arthritis.
- **Supramalleolar osteotomy (SMOT):**
  - **Indication: asymmetric (varus or valgus) ankle OA with partially preserved (at least 50%) joint surface** (**Takakura 2-3A**), **younger patients**.
  - **Medial opening-wedge (or lateral closing-wedge) tibial osteotomy for varus OA** (**overcorrect slightly to about 2-4 degrees valgus**) ± **fibular osteotomy**; **medial closing-wedge for valgus OA**.
  - **Contraindications: end-stage panarthritis**, **severe hindfoot instability**, **neuropathy**, **inflammatory arthritis**, **severe talar tilt (incongruent joint with over about 10 degrees tilt: relative)**.
  - **Delays fusion/TAR** (about 10 years in many patients).
- **Distraction arthroplasty (circular frame distraction 5 mm for about 3 months)**: **young patients**; cartilage repair; variable results.
- **Osteochondral allograft (total ankle allograft)**: high failure; limited role.

11. Ankle arthrodesis

- **Gold standard for end-stage ankle arthritis** (especially **young, heavy laborers, infection history, neuropathy, severe deformity, poor bone stock (talus), failed TAR**).
- **Position: neutral dorsiflexion (90 degrees), 5 degrees hindfoot valgus, external rotation about 5-10 degrees (equal to the**

contralateral foot), talus translated slightly posteriorly (decreases lever arm/knee hyperextension); **equinus position = genu recurvatum/knee pain; varus = lateral foot overload**.

- **Techniques: arthroscopic arthrodesis** (**faster union (about 8-12 weeks), lower complications, less blood loss** in mild deformity: **under about 10-15 degrees**), **open (anterior or transfibular lateral approach)** for deformity/bone loss; **fixation: 2-3 crossed cannulated cancellous screws (6.5-7.3 mm)** ("**home-run screw**" from the posterior malleolus into the talar neck), **anterior plates**, **TTC nail (when subtalar also fused)**.
- **Union rate about 85-95%**; **non-union about 5-15%** (**smoking (strongest modifiable risk), diabetes, AVN of the talus, open approach, infection, previous fracture**).
- **Long-term: adjacent joint arthritis (subtalar about 50-100% radiographically, talonavicular, midfoot) at about 7-20 years**; gait: **reduced sagittal motion compensated by midfoot (about 30% of ankle motion remains through adjacent joints)**; **shoe wear: rocker sole**.
- **Fusion takedown/conversion to TAR** for painful adjacent joint disease (selected).

12. Total ankle replacement (TAR)

- **Modern designs: 2-component (fixed-bearing: INBONE, Infinity, STAR (US: some versions), Salto Talaris)** and **3-component (mobile-bearing: STAR (Europe), Hintegra, Mobility (withdrawn), BOX)**.
- **Indications: older (over about 50-55), lower-demand patients with end-stage OA**, **bilateral ankle arthritis**, **adjacent hindfoot/midfoot arthritis or fusion (TAR preserves motion: protects adjacent joints)**, **inflammatory arthritis**, **good bone stock and soft tissues**, **correctable deformity (under about 10-15 degrees coronal; up to 20 degrees with adjunctive procedures)**.

- **Contraindications: active or recent infection**, **Charcot neuroarthropathy/neuropathy**, **AVN of the talus (large)**, **severe PAD**, **severe uncorrectable deformity/instability**, **poor soft-tissue envelope**, **young heavy labourers (relative)**, **non-reconstructable ligaments**, **paralysis**.
- **Outcomes: survivorship about 70-90% at 10 years** (registry: **revision rates higher than hip/knee arthroplasty**); **functional outcomes comparable to arthrodesis (TARVA trial, UK 2022: similar AOFAS/MOXFQ at 1 year; TAR slightly better in some subgroups)**; **better gait, preserved motion, less adjacent joint degeneration**, but **more reoperations**.
- **Complications: wound complications (commonest; anterior approach: about 5-15%; thin skin)**, **intraoperative malleolar fracture (medial malleolus)**, **aseptic loosening/subsidence (talar component commonest)**, **periprosthetic osteolysis/cysts**, **gutter impingement (medial/lateral; heterotopic ossification)**, **polyethylene wear/fracture**, **infection**, **nerve injury (superficial and deep peroneal)**, **edge loading (malalignment)**.
- **Failed TAR: revision TAR (custom/stemmed talar components)** or **conversion to arthrodesis (TTC nail with bulk allograft (femoral head): "salvage arthrodesis")**: **union rates lower (about 80%)**.
- **Adjunctive procedures to balance the TAR: deltoid release (varus)**, **lateral ligament reconstruction**, **calcaneal osteotomy**, **subtalar/TN fusion**, **gastrocnemius recession**, **SMOT** (staged).

13. Osteochondral lesions of the talus (OLT) (link)

- **Medial (posteromedial: deeper, cup-shaped, often atraumatic: commonest location)** and **lateral (anterolateral: shallow, wafer-like, traumatic)**.
- **Berndt and Harty (radiographic: I compression, II partial detachment, III complete detachment, non-displaced, IV displaced fragment)**, **Hepple (MRI)**, **Ferkel (CT)**.

- **Treatment: non-operative (3-6 months) for non-displaced**; **arthroscopic debridement + bone marrow stimulation (microfracture)** for **lesions under about 1-1.5 cm² (or under 15 mm diameter)**; **fixation of large acute fragments**; **osteochondral autograft (OATS from the knee) or allograft**, **ACI/MACI** for **large (over 1.5 cm²) or cystic lesions or failed microfracture**; **medial malleolar osteotomy** for posteromedial access (**malunion/non-union risk**).

Exam pearl: **ATFL weakest, taut in plantarflexion, intracapsular, injured first; CFL extracapsular, under the peroneals, taut in dorsiflexion, crosses ankle and subtalar joints. Ottawa rules: malleolar bone tenderness at posterior edge/tip of distal 6 cm or can't bear weight 4 steps.**

Exam pearl: **Chronic instability: rehab 3-6 months first; modified Broström-Gould (inferior extensor retinaculum) gold standard; correct cavovarus (else failure); superficial peroneal nerve commonest nerve injury. Syndesmosis: clear space over 6 mm (most reliable), ER stress test most sensitive; anatomic fibular reduction; suture button vs screw.**

Exam pearl: **Ankle OA mostly post-traumatic. Asymmetric OA with over 50% joint preserved = SMOT. Arthrodesis position: neutral, 5 valgus, 5-10 ER, posterior talus; smoking = non-union; adjacent subtalar arthritis long-term. TAR: older, bilateral, adjacent fusion, inflammatory; contraindicated in infection, Charcot, talar AVN; wound problems commonest; TARVA similar to fusion.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.